

A Scenario-Knowledge-Driven Pipeline for Just-in-Time Assistance

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Abstract—Detecting a silently struggling kiosk user is only the first step; deciding whether, when, and how to help depends on scenario knowledge usually buried in model weights and thresholds. We propose a scenario-knowledge-driven pipeline: a single scenario knowledge document, human-authored and version-controlled, configures sensing, constrains LLM reasoning, and shapes a graded intervention proposal. Narration, assistance-need assessment, and proposal are kept separate for independent audit. As proof of concept, we replay two recorded kiosk sessions offline, chosen before the runs for their struggle evidence and retrospective detail. Both cases support what the design promises: checkable reporting and measured escalation. Across 95 updates, every sentence of the append-only narration cites the primitive events underlying it, and the rule layer detects 12 of 13 and 7 of 7 annotated struggle episodes under a strict criterion. The assessor de-escalates on recovery and reaches the top rung exactly once, under maximally converging evidence. At the decisive help-seeking turn, narration, assessment, and the participants’ retrospective accounts converge. The appropriateness of these interventions, the pipeline’s restraint on sessions without struggle, and the document’s transfer to a new scenario frame the agenda.

Index Terms—nonverbal cues, proactive assistance, large language models, human–robot cooperative intelligence

I. INTRODUCTION

Public self-service kiosks (ticketing machines, transit terminals, government service counters) assume users can complete their task unaided. Many cannot, and a large share of those who struggle never explicitly ask for help: they hesitate, repeat the same step, glance around for a staff member, sometimes abandon the task, all without a word. We call these users *silent strugglers*. Recognizing them before they give up enables just-in-time assistance: support at the moment of need and receptivity, not after visible failure or on a fixed schedule [2].

Our prior project, JITAS, asked whether silent struggling at a public ticket kiosk is visible to a machine before the user says anything. From synchronized multimodal recordings, we reported that help-seeking intention is signaled by the *temporal variability* of facial expression, the way the face changes over time rather than any single configuration [1]. The nonverbal signal exists and a machine can read it; that alone does not tell a system what to do.

Detection is only the first step. Knowing that a user may need help does not decide whether, when, or how to act. A poorly timed or overbearing offer can be as harmful as none. It can interrupt a coping user or erode the autonomy

of one who is not. What makes these decisions workable in deployment is scenario-specific knowledge fixed at design time: task structure, user population, salient signals, plausible assistance-need states, available interventions, what counts as socially acceptable help. Each assumption may be sensible; the trouble is it lives *implicitly*, across weights, thresholds, and hard-coded branches, beyond inspection, review, or transfer.

This paper asks what becomes possible when that knowledge is explicit. We propose a scenario-knowledge-driven pipeline in which a single architecture reads a version-controlled *scenario knowledge document* and uses it to configure sensing, constrain LLM reasoning, and shape intervention selection. The document is human-authored and inspectable; it specifies task structure, observable signals, plausible assistance-need states, graded intervention options, and social norms. By design, the system keeps three outputs distinct: an incrementally updated observation narrative, an assistance-need assessment, and a graded intervention proposal, so that every inference step can be audited on its own terms. Our contributions are: (i) a framework that treats scenario knowledge as an explicit, replaceable configuration for adaptive assistance from nonverbal cues; (ii) a first instantiation as a multi-timescale pipeline built around that document as its central artifact; and (iii) a preliminary kiosk case study in which each observation sentence cites the primitive events underlying it, demonstrating the detection-to-decision chain end to end. The *appropriateness* of a chosen intervention is beyond this paper’s scope.

II. RELATED WORK

Assistance-need from nonverbal cues. In intelligent tutoring, facial action units and head pose estimate engagement and help-seeking so hints arrive on time [13]. In socially assistive robotics, gaze fused with language cues decides when to assist with minimal task modeling [14]. Our own prior work detects help-seeking intention at a public ticket kiosk from the temporal variability of facial expression [1]. These efforts share the logic of just-in-time adaptive intervention [2]. Most, however, stop at detection, leaving open how to act on an uncertain need estimate under scenario-specific constraints.

LLM- and VLM-driven proactive interaction. A second line uses large language and vision-language models to decide when and how to act. Closest in spirit, Attentive Support’s LLM-based robot decides when to help a group

and when to stay silent to avoid disturbing it [3]. LaMI regulates multimodal robot behavior through natural-language guidance instead of hand-built state machines [4]. Our scenario knowledge document also selects which detectors run, bounds the inference space, and is frozen under version control. The document, not the code, is thus what a new scenario replaces. MERGE gates an expensive vision-language model behind a lightweight change detector [5]; our instantiation (Section III) gates LLM calls behind rule-emitted events in the same spirit. Further mechanisms include distilled vision-language knowledge for proactive utterances [7], procedure tracking that contextualizes assistance [8], streaming narration of egocentric video into dialogue [9], on-demand sensory context [10], and sensor-language representations [11]. None, however, offers a reusable public-service stack or treats scenario knowledge as an explicit, inspectable, swappable artifact.

Perceptual primitives and appropriateness. Task-specific perceptual primitives (reusable detectors) can be selected from a pool; end-to-end gaze-target detection suits kiosk settings [6]. Beyond accuracy, proactive assistance raises appropriateness questions: unsolicited help can undermine agency or become surveillance, and reasonable deviation can be misread as error [12]. These concerns motivate graded, auditable, explicitly configured intervention rather than maximal proactivity.

III. SCENARIO-KNOWLEDGE-DRIVEN PIPELINE

Fig. 1 shows the architecture. An encoder must turn continuous nonverbal streams (gaze, posture, facial action units, operation traces, the user’s position relative to the machine) into auditable, incrementally updated observation descriptions rather than opaque feature vectors. The document must serve three roles at once: select which detectors run, bound what the LLM can infer, and guide intervention choice. Perception and reasoning must run together within a live interaction’s latency budget. Separating these three concerns lets us ask, for any new scenario, what can be reused, what reconfigured, and what must be rebuilt.

The scenario knowledge document writes out the assumptions buried in code. Our first schema, `kiosk_v1`, has five sections: *task structure*, *observable signals*, *plausible assistance-need states*, *intervention options* (L0–L3), and *social norms*, plus a version stamp (Fig. 1). Structured sections determine what the encoder extracts and bound what the LLM may infer; free-text norms shape intervention choice.

Whether a new setting is an edit or a rebuild depends on which sections change. Edits reach the LLM prompts immediately, and the encoder swaps in a different perceptual primitive, for instance a gaze-target detector [6]. Another self-service kiosk would rewrite the task structure and the social norms while keeping the observable signals and the L0–L3 ladder. Its thresholds would need recalibrating on recordings from that setting. A setting with no screen sequence leaves little task structure standing and needs new perceptual primitives. We have not run such a transfer, so its cost is unmeasured.

The intervention section defines a graded ladder: **L0**, continued silent observation; **L1**, a passive cue making help visible without addressing the user; **L2**, a light, easily declined verbal inquiry; and **L3**, a contextual hint referencing the current step. The ordering follows social exposure, since a contextual hint reveals close observation of the user’s specific struggle, whereas L2 is a generic offer. Direct takeover or staff handoff is beyond L3, outside the automated ladder by design. Choosing a rung depends on how certain the need is and on the scenario’s social norms. In this paper we treat L0–L3 as a design space.

To keep inference inspectable, the system separates the three outputs at every update: observation narrative, assistance-need assessment, and intervention proposal. Kept separate, each is independently auditable, and a confident-sounding proposal cannot hide a weakly supported observation.

Three deployment routes trade off cost, responsiveness, and audibility. **R1** pairs an explicit lightweight encoder with an LLM reasoning over its symbolic output. **R2** maps signals to descriptions or decisions with an end-to-end multimodal model, possibly distilled [7]. **R3**, hybrid and event-triggered, keeps costlier perception and reasoning idle until an event triggers them. In R1 the cheap encoder runs continuously and only the LLM calls are gated; R3 additionally gates perception itself.

This paper instantiates **R1** with multi-timescale event triggering. The encoder is a cheap, deterministic rule layer (M1) running at sensor rate, emitting primitive events that carry provenance to their source signals. An event-triggered LLM narrator (M2) appends to the narrative only when new events justify it; every sentence must cite the primitive events it summarizes (e.g., [E07]). A salience-triggered LLM assessor (M3) then reads the frozen `kiosk_v1` document to produce the assistance-need assessment and an illustrative graded intervention proposal. When no salient event batch arrives, a periodic fallback triggers it every third narration update. Reserving LLM calls for events and salient moments, rather than captioning every frame, keeps the pipeline responsive and its narration traceable. Section IV reports a preliminary case study of this instantiation.

IV. PRELIMINARY CASE STUDY

This preliminary case study asks whether the R1 instantiation runs the detection-to-decision chain end to end with auditable narration. The data come from our earlier JITAS study [1], in which participants booked tickets at a self-service kiosk; the harder of two task variants added friction. Synchronized facial action units, gaze, posture, and interface operations were recorded throughout. In a retrospective interview afterwards, each participant rewatched the recording, marked events, and rated struggle and willingness to receive help. We study two of these sessions, selected before any run for rich pre-request struggle evidence and detailed retrospective accounts. Case A is dense and shorter; Case B is sparser and far longer. Both participants eventually requested help; our analysis window is the pre-request phase. Sessions replay offline (not a deployed

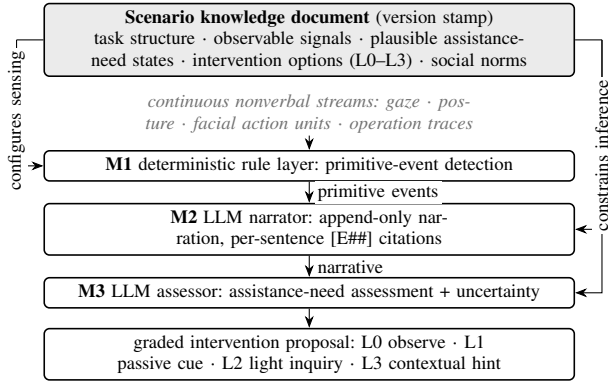


Fig. 1. A version-controlled scenario knowledge document configures sensing (M1) and constrains LLM reasoning (M2, M3), shaping a graded intervention proposal. R1 is the route instantiated here; R2 (end-to-end multimodal) and R3 (gated perception) are alternatives (Section III).

TABLE I
PER-CASE SUMMARY (LOCKED PIPELINE, FROZEN SCENARIO KNOWLEDGE DOCUMENT). M1 = RULE LAYER. DETECTION ALIGNMENT = ANNOTATED/DETECTED/NARRATED (TYPE-MATCHED, ONSET WITHIN 5 S); FIXED-CADENCE BASELINE = HYPOTHETICAL 30 S TICK.

	Case A (dense)	Case B (long)
Time to verbal request (s)	430	865
M1 events (tap bursts)	68 (14)	103 (13)
Narration updates (LLM calls)	33	62
Uncited sentences	0	0
Assessor calls (salient/periodic)	23 (22/1)	33 (26/7)
Fixed-cadence baseline (30 s)	15	29
Interventions L0/L1/L2/L3	4/7/12/0	3/13/16/1
Need none/mon./likely/urgent	4/0/18/1	2/4/26/1
Median narr./assr. latency (s)	4.35/20.75	3.93/22.41
Tokens (fresh in/out)	81,903/35,384	138,585/56,269
Detection alignment	13/12/12	7/7/7

system), each event at its recorded timestamp; LLM latencies are measured but do not gate the replay. Timestamps count from recording start; reported durations end at the participant’s verbal help request, with sensing running briefly past it.

The scenario knowledge document was authored from the interface design, protocol, and two non-case reference sessions, then frozen under version control before any case run. Sessions were recorded under the original study’s informed-consent protocol [1]. Annotations and retrospective interviews were used only for evaluation and never entered any LLM prompt. Participants are pseudonymized, and only feature-level, de-identified text is processed. After the first runs exposed a tap-logging gap (Section V), the detector was corrected and both cases re-run once.

Only run-specific settings differ from Section III. The deterministic rule layer (M1) emits nine primitive event types under fixed, version-controlled thresholds: screen transitions and revisits, operation stalls, long inactivity, tap bursts, scroll thrashes, gaze aversions, posture shifts, and facial-expression-variability spikes (stall: eight seconds idle; tap burst: four presses in two seconds; five-second per-type refractory period). The narrator (M2) and assessor (M3) are unchanged. The reasoning model is Claude Opus 4.8 (claude-opus-4-8), adaptive reasoning, no sampling-parameter overrides; reported latencies are API-side.

Across 95 narration updates and 56 assessments there were

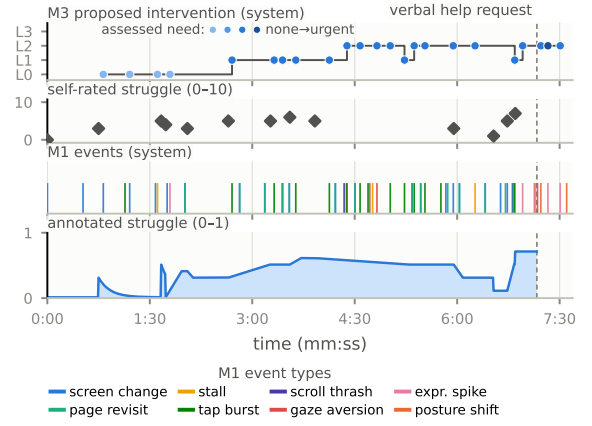


Fig. 2. Case A timeline (mm:ss from recording start). Top to bottom: the M3 assessor’s proposed intervention (L0–L3) as a step trace, each assessment call a dot colored by the assessed need level (light none → dark urgent); the participant’s retrospective self-ratings of struggle (0–10; higher = closer to giving up); M1 events, colored by type (absent types omitted from the legend); and the struggle curve (0–1) from the original study’s annotation procedure [1]. The dashed vertical line marks the moment of the participant’s verbal help request.

zero uncited sentences and no retries or empty responses. Under Table I’s strict criterion, M1 detected 12 of Case A’s 13 annotated episodes and all 7 of Case B’s, each in the final narrative. The zero-uncited result shows only that the narrator followed its format; whether the cited events actually support their sentences was not audited beyond the divergences below. Event triggering also cost more calls than a fixed 30-second tick would have, since both sessions are dense by selection (Table I).¹

Case A (Fig. 2) holds none/L0 (need level/proposed rung) through the smooth opening minutes, turns likely/L1 at 162 s (02:42) on a tap burst at an already-revisited screen. From 263 s it sustains an L2 plateau that begins nearly three minutes before the verbal request. The only *urgent* in Case A (439.7 s) comes about ten seconds after the request. The final posture, gaze, and expression cluster is the request behavior itself. Even there the proposal stays at L2 (L3 “would over-intervene given remaining uncertainty”) and then drops back to likely/L1 at session end. Because of context weighting, an earlier burst at 68 s (01:08), just before a forward transition, is de-weighted; the 162 s burst reads as frustration evidence.

Case B builds more slowly; none/monitor gives way to likely at 219 s, then an L2 plateau backed by current evidence. At 798 s, 67 s before the verbal request, the evidence converges most strongly: three tap bursts and four scroll thrashes over the preceding minute, a posture shift and gaze aversion at 781 s (13:01) seventeen seconds earlier. At that peak the assessor issues the only top-rung escalation in either case, an L3 contextual hint at the document’s gated ticket-amount step, framed as “one gradual step above the L2 the sustained struggle alone would warrant.” In the post-request tail, it de-escalates to monitor/L1 at 906 s as ticket confirmation completes the booking.

¹Token counts are fresh input and output; a constant prompt prefix is additionally cached across calls.

The separated outputs can be checked against the participant’s own retrospective account. At the decisive turn, all three streams coincide: the narrative’s “turning toward the surroundings for help” at 781 s (13:01), the assessor’s sole L3 at 798 s (67 s before the request), and the participant’s resolution, “I’d pretty much exhausted everything and still couldn’t get through, so I made my peace with it: one more try, and if that failed, I’d call for help” (translated from Japanese). All 24 participant-marked events are accounted for (author-coded, single-coder, unblinded): 9 agree (same behavior described), 11 partial (related, missing the specific quality), 2 diverge (contradicted), 2 out of scope (no behavioral trace). When the participant recounts, “Even though I’d set the seats for the number of passengers, it still wouldn’t let me move on, and I was completely at a loss” (translated from Japanese), the narrative independently reports a tap-burst run on the seat-selection screen.

The misses are as informative. Both divergences fall in Case A and share one failure mode, a struggle-biased narrator with no positive-affect primitive. The participant recalled two moments of relief: “I spotted the ‘Information’ button and pounced on it ... and felt happy” (translated from Japanese), then the final advance to the next page. At both moments the narrative keeps reading “sustained cycling ... rather than advancing.” The screen changes were captured, so these failures are interpretive, not sensory. Two further edge cases remain. A few early Case B assessments read likely while annotated struggle was near zero (over-sensitive, though neither case reverses the rank order). Case A’s one undetected episode is an artifact of the criterion boundary: the same tapping fired six seconds before the annotation window and the five-second refractory period suppressed a second event.

Every reading can be traced back to its signals. The Case B sentence at 781 s cites [E83, E84], which the event log resolves to a posture-shift and a gaze-aversion rule; M1 fired both deterministically from signal windows ending at that timestamp, under the five-second refractory period. The interpretive sentence, the assessment citing it, and the raw signal form one traceable chain.

V. DISCUSSION AND RESEARCH AGENDA

Building the pipeline taught transferable lessons. First, auditability came by construction. An early full-rewrite narrator drifted into summarizing only recent events late in sessions; appending only new, cited sentences eliminated it and left an append-only log. Second, memory changes conclusions. With identical events and prompt, keeping the full narrative history in the assessor’s context pushed it from 11 to 18 of 21 “likely” verdicts and from zero to 13 light-inquiry (L2) proposals on a non-case reference session. Only recency weighting restored de-escalation and re-escalation as evidence aged. Third, the audit trail proved its value. The first runs missed every rapid-tap episode because the kiosk emits touch events the click-based counter never saw, a gap per-event provenance exposed immediately. After the detector fix and recalibration of the two-second tap window on reference sessions, alignment

moved from 1/13 and 3/7 to 12/13 and 7/7. Fourth, event-triggered cost scales with evidence density (a real cost here, a hypothesized saving on quiet sessions), consistent with gating expensive reasoning behind lightweight change detection [5].

In both cases retrospective willingness to receive help peaked well before escalation from passive cue to light inquiry (L2), by about 100 s in Case A and 240 s in Case B. Need is detected promptly, but escalation is conservative relative to the readiness participants reported. This is a deliberate bias toward the norm of under-intervention in public spaces [12]. Latency adds a second gap. With a median assessment latency of twenty-one seconds, escalations at a live kiosk would arrive after the moment of need, and real-time feasibility is untested. A preliminary check points to the time spent writing the assessment text. Published results on faster decoding and streaming output [15], [16] suggest that this cost can fall substantially, and we will pursue that next.

Four limitations bound what two cases can show. First, the two sessions were chosen before the runs, for struggle evidence and retrospective detail. Both contain sustained struggle, so the pipeline was never asked to stay quiet, and specificity is untestable here. With 44 of 56 assessments at likely and two more at urgent, the convergence claims concern timing and rank rather than discrimination. Second, the appropriateness of the proposed interventions is not evaluated. Third, the two cases show that the pipeline runs and can be audited. Whether making scenario knowledge explicit works better than encoding it implicitly is untested. Fourth, one reasoning model, one frozen threshold set, and single-coder unblinded annotation leave model dependence and coder reliability unknown.

The agenda covers negative-control sessions for specificity; independent document authoring; cross-scenario transfer, separating document replacement from encoder reconfiguration [6]; a controlled comparison of the explicit document against the same knowledge left implicit in prompts or weights; R1 versus R2 and R3 [7]; the deferred appropriateness evaluation; and integration with a service robot or cobot [3].

VI. CONCLUSION

This paper asked what becomes possible when the scenario knowledge behind when-and-how-to-help decisions is made explicit. In an end-to-end two-case study, a single version-controlled document configured sensing, constrained LLM reasoning, and kept every inference step auditable. The narration stayed traceable to its primitive events, and the assessor followed two different struggle dynamics with graded restraint. The appropriateness evaluation with users comes next. Beyond it lie sessions without struggle, transfer to another scenario, and a robot that acts on the proposals.

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